

CONTROL, DEPENDENCE AND CAPTURE

The Sovereign Ledger

Payments, clearing and securities registries in the programmable transition

Global architecture, with Australia as the worked example

Integrated briefing current to 5 August 2026 · approx. 90 minutes

The question is not who moves the last dollar

Who controls the path that determines whether the dollar can reach the final ledger?

Formal authority

- Central bank issues the settlement asset
- Domestic law defines finality
- RTGS records the final debit and credit

Sources: RBA, [RITS overview](#); Bank of England, [RTGS overview](#).

Operational authority

- Identity and admission
- Routing, sequencing and liquidity locks
- Code, keys, data, recovery and exit

Orders of magnitude

- **Swift** moves value equivalent to world GDP every two to three days.
- **CLS** handles FX settlement values on the order of US\$7 trillion on an average day.
- **LCH SwapClear** clears about 95% of vanilla OTC interest-rate swaps globally, in 28 currencies.
- **Indeval's DALÍ** settles over US\$250 billion of Mexican securities daily.
- **Kinexys** (J.P. Morgan) reports over US\$4 trillion processed since inception.

These layers govern flow, not stock. Small design choices compound across these volumes daily.

Sources: Swift, July 2026 ledger announcement; CLS; LSEG SwapClear; Indeval/INFORMS; J.P. Morgan Kinexys.

Control of a settlement system is an eight-layer capability

- | | | | |
|---|-------------------------------------|---|---|
| 1 | Law and finality | 5 | Code, keys and specialist knowledge |
| 2 | Rules, access and pricing | 6 | Connectivity, identity and certificates |
| 3 | Authoritative money or title record | 7 | Data, routing and optimisation |
| 4 | Live operations and recovery | 8 | Credible exit and substitution |

A jurisdiction can retain layers 1–3 while conceding layers 4–8 to external vendors and international utilities.

One payment contains several control events

Initiate → authenticate → route
→ clear / optimise → reserve liquidity
→ **settle in central-bank money**

The final RTGS entry can remain publicly operated even when every economically important choice was made upstream.

Sources: BIS CPMI, payment-system concepts; RBA RITS; Swift shared-ledger publications.

Tokenisation adds records without removing law

Four records are called “the ledger”

- Central-bank money ledger
- Securities or fund ownership record
- Commercial-bank liability ledger
- Coordination / orchestration ledger

The decisive questions

- Which record is legally authoritative?
- Who can upgrade or pause it?
- What happens in insolvency?
- Can the market operate without it?

Sources: BIS Project Agorá final report; central-bank and FMI rulebooks surveyed in the companion report.

The central risk is a private constitution

A shared technical layer can become the place where market access, liquidity and permissible behaviour are actually governed.

- No conspiracy is required: institutions rationally join every credible network.
- Network effects make the chosen identity, token and messaging standards difficult to escape.
- The result can resemble coordination because the same firms occupy every architecture.

Sources: Ownership and participation analysis in The Sovereign Ledger; public corporate and FMI disclosures.

Roadmap, method and a disclaimer

- | | | | |
|-----|----------------------------------|------|--------------------------------|
| I | What the core already looks like | V | Swift and competing networks |
| II | Australia's current position | VI | Technology and company lineage |
| III | Eight Agorá jurisdictions | VII | ASX: failure and consequence |
| IV | Agorá and its lineage | VIII | The test for any jurisdiction |

Method: every assessment is tagged **documented**, **inference** or **unproven**.

This is not an argument for autarky: shared infrastructure is desirable — on inspectable, substitutable, contestable terms.

PART I

What the core already looks like

Payments, clearing, title and programmable finance

FMI ownership follows four recurring models

Model	Examples	Primary capture risk
Public operator	RITS, Fedwire, RT2	procurement and capability dependence
User-owned utility	Swift, CLS, SIX, TCH	incumbent admission and rule-book power
Listed FMI group	ASX, LSEG, Deutsche Börse, TMX	shareholder returns over resilience
Venture-backed platform	Digital Asset, R3, Fireblocks	vendor, investor and exit concentration

Sources: Operator governance and shareholder disclosures cited in the companion report.

Securities control is not exchange ownership

Trading venue

- Matches buyers and sellers
- Can be competitively duplicated

CCP

- Novates and nets obligations
- Controls margin and default management

CSD / registry

- Establishes or supports legal title
- Coordinates securities settlement
- Much harder to substitute in crisis

Sources: CPMI-IOSCO PFMI; ASX CHESS; Euroclear CREST materials.

Commercial-bank money already settles at the core

- **Euroclear Bank**, the world's largest international CSD, settles the cash leg in commercial-bank money: claims on Euroclear Bank itself.
- **CLS**: funding and defunding move through central-bank RTGS accounts, but settlement occurs across accounts on the books of CLS Bank; CLSNet nets outside central-bank money.
- **Correspondent banking** discharges cross-border obligations in commercial-bank money by construction.

**The pure central-bank-money story was already porous.
Tokenisation does not introduce commercial-bank settlement money;
it makes it programmable.** **documented**

Message-layer control is demonstrated power

- 2022: EU sanctions disconnected major Russian banks from Swift — the messaging layer used, openly, as a policy instrument.
- CIPS (2015) and the e-CNY are the most developed state-built responses; mBridge (Part IV) is the multi-central-bank variant.
- Every jurisdiction absorbed the same lesson: whoever operates the coordination layer can exclude.

No further proof is required that the layers above final settlement carry decisive power. The only open question is who holds them next. **documented**

Sources: EU Council sanctions decisions (2022); PBoC CIPS materials; BIS mBridge documentation.

The physical layer concentrates as well

- FMI and their vendors increasingly run on a small set of hyperscalers and shared software supply chains — layers L4 and L6 in practice.
- 19 July 2024: a single faulty CrowdStrike update simultaneously disrupted airlines, banks and exchanges worldwide — one vendor, global propagation.
- Cloud regions, HSM fleets, certificate authorities and privileged tooling sit outside most jurisdictions' practical reach.

**Layer analysis must include hosting, keys and endpoint software —
not only rulebooks. documented**

Sources: Public incident reporting, July 2024; FSB third-party dependency workstream.

PART II

Australia's current position

Payments, clearing, title and programmable finance

Australia owns the monetary core

- RBA owns and operates **RITS**
- ESA transfers provide final wholesale AUD settlement
- FSS supplies 24/7 central-bank settlement for NPP
- CHES remains the legal-title and equity-settlement spine

Australia issues, records and finally settles its own currency. That does not by itself confer end-to-end operational control.

Sources: RBA, [RITS](#); RBA, [FSS](#); ASX, CHES settlement.

Australia depends above the final ledger

Layer	Important provider	Dependency	Tags
High-value messaging	Swift	standards, connectivity, certificates	L6
Retail fast payments	AP+ / Swift / FSS	scheme, gateway and settlement integration	L2 L6
COIN connectivity	TNS	network operations and support	L6
Equity replacement	TCS BaNCS	foreign product and specialist capability	L5
Future orchestration	undecided	routing, locks, data and workflow authority	L2 L7

Sources: RBA, AusPayNet, AP+, Swift, TNS, ASX and TCS disclosures.

The largest concession already happened

- The large majority of AUD OTC interest-rate derivatives clears at **LCH SwapClear** in London (~95% of vanilla OTC swaps globally).
- The CFR's 2012–2015 location-policy debate settled on overseas clearing under safeguards (licensing, RBA standards, Bank of England cooperation).
- ASX Clear (Futures) offers a domestic AUD/NZD OTC service; the global liquidity pool sits offshore.

Margin, default management and crisis decisions for the core AUD rates market already sit with a London CCP. The tokenisation debate starts here, not from autonomy. **documented**

Sources: CFR, [OTC derivatives reports](#); RBA Bulletin (2018); LSEG, [SwapClear](#); ASX OTC clearing.

RITS modernisation is the live decision

Opportunity

- Modern interfaces and data
- Longer operating hours
- More settlement in central-bank money
- Multiple synchronisers, not one franchise

Present weakness

- Three PFMI principles partly observed
- Core Modernisation delays
- Congested change portfolio
- Scarce specialist capacity

Sources: RBA, 2026 ratings and material developments.

PART III

Eight Agorá jurisdictions

Payments, clearing, title and programmable finance

The euro area: public anchor, private machinery

Existing stack

- T2 / T2S / TIPS central-bank services
- Euroclear and Clearstream post-trade groups
- Eurex and LCH clearing power

Tokenisation route

- 2024 trials: Bundesbank Trigger, Banque de France DL3S, TIPS Hash-Link
- **Pontes** bridges market DLTs to TARGET;
Appia is the integrated long run

The euro's monetary anchor is public; much of its collateral and securities machinery is not.

Japan separates bank money from orchestration

- BOJ-NET settles central-bank money and JGBs; Zengin-Net clears retail/interbank payments.
- DCJPY is live with one issuing bank: GMO Aozora.
- DCP transmits instructions; the issuing bank's Financial Zone record is characterised as the deposit.
- Multi-bank interbank settlement remained an FSA-supported proof of concept in April 2026.

Public evidence does not establish Article 73 insolvency protection for DCP-level netting. **inference**

Sources: BOJ; DCP legal/product disclosures; Japan Payment Services Act arts 64, 71, 73.

Korea is testing tokenised deposits under a public core

- BOK-Wire+ remains the high-value central-bank-money system.
- The Bank of Korea has explored wholesale CBDC and deposit-token architectures.
- Six Korean banks in Agorá provide broad domestic coverage: Hana, IBK, KB, NongHyup, Shinhan and Woori.
- The policy question is whether a shared platform complements BOK-Wire+ or becomes the indispensable access layer above it.

Sources: Bank of Korea speeches and Agorá participant package.

Mexico tests cross-border reach without surrendering SPEI

- Banco de México operates SPEI and the account system SIAC; SPID handles USD payments between legal entities.
- **Indeval**, the private CSD, operates the DALÍ securities-settlement system, linked to Banco de México's systems.
- Banorte supplies a domestic bank perspective; Monex adds specialist FX and direct CLS participation.
- BBVA connects the peso domain to a global group and published Agorá validation work.

The test is whether domestic peso discharge remains anchored to Banco de México money.

Switzerland operates both public and private models

- SIC is operated by SIX under the SNB mandate.
- SNB wholesale CBDC has settled live transactions on SDX since 2023.
- Helvetia now compares integrated wCBDC with a production RTGS link.
- Fnality-like private settlement and multi-bank deposit tokens remain legally distinct from SNB money.

Switzerland is testing architecture without pretending every Swiss-franc token is the same liability.

Sources: SNB Project Helvetia; SIX/SDX; Swiss multi-bank deposit-token PoC.

The UK renewed the core and made synchronisation a role

- The Bank of England's **RT2** went live in April 2025.
- Mastercard's VocaLink runs key retail infrastructure; Euroclear UK & International operates CREST.
- The Synchronisation Lab formalises external operators coordinating RTGS with asset ledgers.
- Fnlity supplies reserve-backed settlement tokens; Lloyds and NatWest link deposit-token work to Agorá.

A synchronisation operator need not own RTGS to control when an instruction is released. Its power lies in conditions, timing and earmarks.

The United States privatises dollar governance

- Fedwire and FedNow provide public settlement rails.
- TCH owns CHIPS, RTP and EPN on behalf of the large US commercial banks that own it.
- DTCC controls the central US securities post-trade chain.
- Kinexys, Citi and BNY operate proprietary commercial-bank-money ledgers.

The dollar's monetary anchor is secure; access to programmable dollars may still be governed through private tollbooths.

Sources: Federal Reserve; TCH; DTCC; JPMorgan, Citi and BNY token-service disclosures.

Canada combines a public core with imported optimisation

- Payments Canada owns Lynx; settlement occurs in Bank of Canada money.
- TMX owns CDS and CDCC.
- TMX and Clearstream jointly operate the Canadian Collateral Management Service.
- Canada joined Agorá in May 2026, after the seven-currency prototype.

CCMS shows how a foreign-controlled optimiser can shape domestic repo without owning the central bank or CSD.

Sources: Payments Canada; Bank of Canada Agorá announcement; TMX/Clearstream CCMS.

Eight jurisdictions, eight constitutional positions

Area	Public anchor	Tokenisation path	Main exposure	Tags
Euro	TARGET	Pontes / Appia	private post-trade concentration	L3 L7
Japan	BOJ-NET	DCJPY / sandbox	multi-bank discharge	L1 L3
Korea	BOK-Wire+	CBDC/deposit tests	access-layer dependence	L2 L8
Mexico	SPEI	Agorá / bank tokens	peso settlement anchor	L1 L3
Switzerland	SIC / SNB wCBDC	Helvetia / SDX	SIX and platform control	L2 L4
UK	RT2	sync / Finality	external operator and earmarks	L4 L7
US	Fedwire	TCH / Kinexys / DTCC	private dollar governance	L2
Canada	Lynx	Agorá / CCMS	imported collateral optimisation	L7

Sources: Central-bank and FMI sources summarised on preceding slides.

PART IV

Agorá and its experimental lineage

Payments, clearing, title and programmable finance

Agorá joins bank deposits to reserve domains

Tokenised commercial-bank deposits

↓ shared payment contracts and atomic conditions ↓

Separate currency-specific central-bank reserve domains

- Seven original currency areas; Canada joined in May 2026 after the completed prototype.
- Besu-based experimental architecture with ISO 20022 and legal-entity identity.
- July 2026 real-value phase: 17 scenarios, about CHF800,000 aggregate value, roughly 80 seconds average initiation-to-settlement.

Sources: BIS, [Agorá final report](#) (27 May 2026); BIS, [real-value testing summary](#).

Agorá proved mechanics, not a constitution

What it demonstrated

- Atomic multi-currency transactions
- Shared programmable workflows
- Technical operation across institutions
- Real-value test execution

What remains open

- Production owner and operator
- Rulebook and legal designation
- Loss allocation and insolvency
- Validator control, recovery and exit

Sources: BIS Agorá final report and real-value testing documentation.

The common ledger does not erase RTGS

- Reserve tokens must be funded from, and redeemable into, the relevant central-bank accounts.
- Commercial-bank tokens remain claims on named banks.
- Cross-currency atomicity can occur on the shared platform while domestic finality remains anchored to each jurisdiction's money.
- A production design could nevertheless make the common layer operationally indispensable.

A currency authority keeps control only if it can isolate its domain and continue settlement without the shared layer.

Sources: BIS Agorá architecture and companion-report legal analysis.

Forty-two firms shaped the prototype

Concentrated capabilities

- Global correspondents and FX dealers
- Japanese and Korean domestic bank clusters
- Custodians and tokenised-deposit operators
- Eurex, Euroclear and SIX Digital Exchange
- Visa, Mastercard and Swift

Under-represented interests

- End users and corporates
- Smaller-bank associations
- Independent competition voices
- Public-interest and civil-society representatives

Sources: BIS/IIF private-sector participation package and final participant list.

Participation is not ownership

The 42 firms were a design and testing coalition—not shareholders in a disclosed production utility.

- Real-value testing was conducted by 28 institutions on behalf of all participants, in transactions of CHF 9,000–125,000.
- Project status therefore proves workstream access, not execution of a real-value transaction — and not endorsement of every design conclusion.
- Only a few specialised roles were uniquely disclosed: Eurex as sole CCP; SIX and Kaleido in later test roles.
- The future operator, ownership and governance remain undecided.

Sources: BIS/IIF application package; BIS Agorá final and real-value testing reports.

Dunbar tested one multi-CBDC platform

- RBA, Bank Negara Malaysia, MAS and South African Reserve Bank.
- Two prototypes: **Corda/R3** and **Quorum/Partior**.
- Direct cross-border transactions in several hypothetical CBDCs.
- Governance, non-resident access and central-bank autonomy were the hard problems.

Dunbar asked whether several national monies could inhabit one platform without a platform super-user overruling the issuers.

Sources: BIS, Project Dunbar final report.

Jura put real central-bank money on Corda

- Banque de France, SNB and BIS with SIX, UBS, Credit Suisse, Natixis, R3 and Accenture.
- Real-value euro and Swiss-franc wCBDC plus tokenised commercial paper.
- Subnetworks and dual-notary signing preserved currency-specific vetoes.

A central bank can retain issuance authority while depending on a third-party platform for availability and software evolution.

Sources: BIS, [Project Jura report](#).

mBridge is the central-bank-built alternative

- Hong Kong, Thailand, China, UAE and later Saudi Arabia.
- Custom mBridge Ledger with a validating node for each founding authority.
- Real-value bank participation; MVP reached in 2024; BIS handed the project to participating authorities in October 2024.
- The handover followed public debate about whether multi-CBDC platforms could route around established payment controls. [inference](#)

mBridge can reduce dependence on correspondent chains and Swift—but validator control does not resolve sanctions, admission or rulebook politics.

Sources: BIS, [mBridge project page](#) and 2022 report.

Three constitutional models are competing

Model	Examples	Concentration danger
Common multi-issuer platform	Dunbar, mBridge	common governance and validator layer
Shared bank/public workflow	Agorá, Swift ledger	incumbent orchestration and identity
Interlinked national ledgers	Cedar-Ubin+, Pontes, Meridian FX, Helvetia	bridge, oracle and synchroniser power

There is no inevitable one-ledger future. The contest is where coordination authority sits.

Sources: BIS Innovation Hub project reports; ECB Pontes; SNB Helvetia; New York Fed Cedar. Project-by-project control questions in the appendix.

PART V

Swift and competing networks

Payments, clearing, title and programmable finance

Swift is moving from messages to commitments

- September 2025: shared-ledger plan designed with more than 30 institutions.
- March 2026: Besu-based MVP records and validates interbank payment commitments.
- Swift operates the ledger; banks retain keys, deposit assets, compliance and ultimate funding.
- July 2026: controlled initial-use phase announced with 17 banks.

Swift is not becoming the settlement asset. It is trying to become the trusted workflow through which settlement is arranged.

Sources: Swift, shared-ledger announcements of September 2025, March 2026 and 9 July 2026.

The 17-bank cohort is globally distributed

Europe / Americas

- BNP Paribas
- BNY
- Citi
- HSBC
- Itaú
- Lloyds
- UBS
- Wells Fargo

Asia-Pacific

- ANZ
- DBS
- MUFG
- OCBC
- UOB

Middle East / Africa

- First Abu Dhabi Bank
- FirstRand
- Mashreq
- Standard Chartered

Sources: Swift, 9 July 2026 initial-use announcement.

JPMorgan is absent from 17, not from the strategy

- JPMorgan was in Swift's earlier design group, but not the July 2026 cohort of 17.
- It is an Agorá participant and owns **Kinexys**.
- Kinexys operates payments, tokenised assets and information networks.
- JPM Coin has operated on proprietary infrastructure and Base; native Canton issuance is planned.

JPMorgan is hedging architectures, not choosing one.

Sources: Swift design and cohort releases; JPMorgan Kinexys/JPM Coin; Canton deployment announcement.

CLS is the mature comparator

- Bank-owned cross-border utility with powerful network effects.
- Payment-versus-payment settlement eliminates principal risk for eligible FX.
- Funding moves through each currency's central-bank RTGS account; settlement itself occurs across accounts on the books of CLS Bank.
- Swift provides connectivity; cooperation is public, not evidence of merger.

CLS shows how a private utility can become indispensable without replacing national central-bank money.

Sources: CLSSettlement, shareholder and third-party materials; Swift/CLS connectivity.

TCH proposes bank-owned tokenised deposits

- June 2026 proposal backed by 17 named financial institutions.
- On-chain clearing and settlement of tokenised bank deposits.
- Intended connectivity to CHIPS and RTP.
- TCH is owned by its large US member banks.

The bank branch of programmable dollars preserves deposit franchises and owner-bank governance.

Sources: The Clearing House, [5 June 2026 initiative](#).

Open USD proposes a distribution coalition

- Announced June 2026 by Open Standard / OnePay; more than 140 businesses named, not yet a live stablecoin at the cutoff.
- Fee-free uncapped mint and redemption proposed; reserve earnings shared with participating distributors after a management fee.
- Context: the GENIUS Act (July 2025) bars paying yield to stablecoin holders — revenue-sharing with distributors is the compliant substitute.

Open USD competes by paying platforms for distribution; TCH competes by preserving bank deposits.

Sources: [Open Standard](#); [OnePay launch announcement](#); GENIUS Act (2025).

Tokenised MMFs become payment-adjacent collateral

- BUIDL, BENJI/FOBXX and USYC combine fund claims with transferable tokens.
- They can accelerate subscription, redemption and collateral mobility.
- They are securities/fund interests—not bank deposits or central-bank money.
- NAV, gates, official ownership records and eligible-investor restrictions remain decisive.

A token can trade like cash operationally while retaining investment and insolvency risk legally.

Sources: Franklin Templeton BENJI; Circle USYC; Securitize/BlackRock BUIDL materials.

ERC standards solve interfaces, not finality

Standard	Function	Does not establish
ERC-20	fungible token interface	legal title or redemption
ERC-4626	synchronous tokenised vaults	investor eligibility or NAV certainty
ERC-7540	asynchronous request lifecycle	settlement asset or insolvency priority
ERC-7575	multi-asset vault extensions	authoritative registry status

Sources: Ethereum Improvement Proposals 20, 4626, 7540 and 7575.

Identity is the quiet gatekeeper

Architectures

- Central registry: one provider owns eligibility data
- Reusable issuer-signed credentials
- Federated multi-authority verification
- Zero-knowledge proofs of eligibility

Securitize, the live instantiation

- Transfer agent, broker-dealer/ATS, fund admin
- Allowlists encode jurisdiction and eligibility
- BlackRock: 2024 round and a board seat
- Listed as SECZ, July 2026 (Cantor-sponsored)

Whoever controls the identity-to-wallet mapping can become the private gatekeeper to tokenised markets.

Sources: W3C Verifiable Credentials 2.0; Securitize corporate and investor releases; SEC filings.

PART VI

Technology and company lineage

Payments, clearing, title and programmable finance

Technology choice is constitutional choice

	Data model	Strongest feature	Capture point
Besu Canton	shared EVM state party-specific contracts	standards and labour pool privacy and cross-app atom- icity	validators, privacy, upgrades synchroniser, validators, Daml
Corda Fabric	shared bilateral states channels / private data	regulated workflows modular enterprise control	notary and network manager ordering and membership service
Public chain	replicated state	reach and composability	validators, bridges, RPC, custody

Sources: LF Decentralized Trust; Digital Asset; R3; Ethereum documentation.

Besu is the institutional common denominator

- Apache-2.0 Ethereum client under LF Decentralized Trust.
- Public Ethereum or permissioned QBFT/IBFT networks.
- EVM compatibility exposes mature tools and token standards.
- Used differently by Agorá, Swift, Finality, DTCC and JSCC.

Open code reduces code lock-in; it does not eliminate control over validators, hosting, privacy services, keys or releases.

Sources: LF Decentralized Trust, [Hyperledger Besu](#); operator disclosures.

Finality proves Besu can settle regulated value

- Sterling Finality uses permissioned Ethereum/Besu.
- Tokens are fully backed through a Bank of England omnibus-account arrangement.
- The system is recognised and designated for settlement finality.
- Twenty-four banks/FMIs publicly described as backing the group.

Production technology does not make the token central-bank money; each currency needs its own legal and central-bank arrangement.

Sources: Finality; Bank of England RTGS/CHAPS annual report and recognition materials.

Digital Asset came from the DRW orbit



- Trading and crypto-market liquidity
- Derivatives workflow and clearing
- Research cryptography

Sources: Digital Asset founder biographies and March 2015 launch; DRW, Don Wilson profile.

Founding record and the Masters era

March 2015 announcement

- Founders: Don Wilson and Sunil Hirani
- Technology described as their brainchild
- Both joined the board

Current corporate history

- Co-founders: Yuval Rooz, Eric Saraniecki, Shaul Kfir
- Masters: CEO from the March 2015 launch; not an original founder
- Reframed DLT as regulated-market infrastructure; led the ASX relationship

Best reconstruction: Wilson/Hirani financial founders; Rooz/Saraniecki/Kfir the operating-technical founding group; Masters supplied institutional access and legitimacy.

inference

Sources: Digital Asset launch release, 2016 financing release and current corporate biographies.

Technology assembled by acquisition

Date	Acquisition	Capability added
June 2015	Hyperledger Inc.	permissioned ledger, team and trademark
June 2015	Bits of Proof	Bitcoin and enterprise-ledger engineering
2015/16	Blockstack.io	private-blockchain services; unrelated to Stacks
April 2016	Elevent	contractual modelling language that became Daml
2023 on-ward	Canton network	interoperable private Daml applications

Hyperledger mark and code were donated to the Linux Foundation (2015–16); Fabric is IBM lineage, Besu is Consensus lineage — Digital Asset owns neither.

Canton should not be projected backwards into the platform ASX selected in 2016.

Sources: Digital Asset corporate milestones; acquisition announcements; Linux Foundation formation and 2016 code-proposal announcements.

The Ethereum lineage is separate

Date	Ethereum	Digital Asset
Nov 2013	white paper	—
Jan–Sep 2014	announced, specified, funded	organised later in 2014
Mar 2015	prototypes	public company launch
Jul 2015	mainnet launch	operating and acquiring teams

Digital Asset predates the Ethereum mainnet, not the Ethereum project. Later overlap came through JPMorgan, Consensus, Hyperledger and shared investors—not a common founding team.

Sources: Ethereum official history; Digital Asset March 2015 launch.

The 2016 coalition changed the company, and kept paying

US\$50m+ strategic investors (2016)

- ABN AMRO, BNP Paribas, Citi, JPMorgan, PNC, Santander
- ASX, CME, Deutsche Börse, DTCC, ICAP
- Accenture and Broadridge

Board representation

- BNP Paribas
- Deutsche Börse
- JPMorgan
- DTCC

2016	US\$50m+	banks, exchanges, CSD/clearing and integrators
2025	US\$135m	Tradeweb/DRW plus dealers, DTCC and crypto firms
2026	US\$355m	a16z, ADIA, banks, brokers, exchanges and crypto investors

700+ Canton ecosystem participants reported June 2026; the 2026 round disclosed no valuation.

Sources: Digital Asset/Broadridge 21 January 2016 financing announcement; 2025 and 11 June 2026 financing releases.

Canton coordinates private applications

- Participants hold party-specific contract views rather than one public payload ledger.
- Synchronisers sequence encrypted messages.
- Mediators coordinate atomic commit across applications.
- Global Synchronizer governance uses Foundation and super-validator voting.

Canton distributes data but can still concentrate operational power in synchronisation, validation, hosting and Daml expertise.

Sources: Digital Asset Canton participant architecture; Canton Foundation governance.

Institutions hedge every architecture

- JPMorgan: Agorá, Swift design, TCH, Kinexys, Partior and Canton.
- DTCC: Digital Asset investor; Canton and Besu deployments.
- Euroclear: Agorá, Fnality and Canton collateral network.
- Visa and Mastercard: Agorá, Swift/Besu groups and Open USD.
- Tradeweb: Digital Asset and Securitize; majority owned by LSEG.

Strategic optionality lets incumbent institutions shape the winner whichever stack prevails.

Sources: Corporate investment, participant and partnership disclosures mapped in the companion report.

PART VII

ASX: failure, conflict and consequence

Payments, clearing, title and programmable finance

ASX made the market a proving ground

- January 2016: exploratory Digital Asset work plus 5% equity stake; June 2016: 8.5% and a board seat.
- 2017: commitment to DLT-based CHES replacement.
- November 2022: project paused; A\$245–255 million written off.
- November 2023: TCS BaNCS selected for a staged replacement.
- 20 April 2026: Release 1 (clearing) goes live; Release 2 (settlement and subregister) is projected toward the end of the decade.

Vendor selection, venture investment, product development and critical-system replacement became inseparable.

Sources: ASX announcements; ASIC inquiry and Federal Court penalty materials.

Four risks were accepted together

Risk	Why it was foreseeable
Technology	no equivalent national production deployment
Integration	acquired teams and components still forming one stack
Governance	ASX was buyer, investor and board participant
Disclosure	admitting failure damaged both infrastructure and investment narratives

“Incompetence” is incomplete: this was a foreseeable institutional-design failure.

inference

Sources: ASX chronology; Digital Asset acquisition/funding history; ASIC findings.

The conflict was real

- ASX had a financial and reputational interest in vendor success.
- CME, Deutsche Börse, DTCC and major banks invested alongside it.
- Those firms gained exposure to reusable post-trade technology and requirements knowledge.
- Digital Asset lacked an equivalent live national-market deployment.

Conflict establishes a duty for exceptional independence. It does not by itself establish corrupt action.

Sources: Digital Asset 2016 investor/board release; ASX disclosures; ASIC inquiry.

The official findings are already severe

- ASIC inquiry: resilience compromised in pursuit of high shareholder returns; governance, capability and stewardship failures identified.
- ASX admitted its 10 February 2022 “progressing well” statement was misleading — contravening ss 12DA and 12DB(1)(a) and (e) of the ASIC Act.
- July 2026: the Federal Court imposed a A\$20.5 million penalty plus A\$3 million towards ASIC’s costs.

The public record establishes conflicted and misleading stewardship of critical infrastructure. **documented**

Sources: ASIC, April 2026 inquiry findings; ASIC, July 2026 CHESS penalty.

The sabotage hypotheses rank differently

	Hypothesis	Assessment
1	governance failure and shareholder capture	documented
2	reckless experimentation in conflicted procurement	documented
3	ecosystem enclosure without central conspiracy	inference, partly evidenced
4	tacit tolerance of failure by beneficiaries	inference, weak
5	coordinated sabotage for a foreign takeover	unproven

Sources: Evidence ledger and counterfactual assessment in the companion report.

Later benefit is not proof of authorship

- Digital Asset gained requirements knowledge, credibility and a flagship narrative.
- Its investors gained option value across post-trade architectures.
- ASX sold its remaining 5.4% stake in 2025 for about A\$57 million.
- ASX reported only about A\$10 million pre-tax gain over acquisition cost—far below the project write-off.

The strategic springboard is credible; a disclosed ASX investment-profit motive is not.

inference

Sources: ASX 2025 sale announcement and project write-off disclosures.

Cboe and TMX did not replace CHESSE

- ASIC introduced competition rules limiting misuse of ASX monopoly power.
- ASIC approved Cboe Australia as a listing venue in October 2025.
- TMX agreed in 2026 to buy Cboe Australia and Canada, subject to approval.
- ASX—not Clearstream or TMX—selected TCS BaNCS and remains the CHESSE operator.

De-monopolisation is real; the claim that Clearstream “swooped in and took CHESSE” is unproven.

Sources: ASIC competition/listing releases; TMX/Cboe transaction; ASX TCS selection.

PART VIII

The test for any jurisdiction

Payments, clearing, title and programmable finance

Dominance without issuing AUD

- Select correspondents and executable FX paths.
- Determine collateral eligibility and substitution.
- Sequence obligations and reserve liquidity.
- Encode identity, sanctions and market-access rules.
- Send only the final net or gross instruction to RITS.

RITS could retain legal finality while becoming the last actuator of decisions made elsewhere.

Sources: Agorá, Swift, CLS, Meridian FX and synchronisation architectures analysed in the companion report.

A CLS-style utility can capture workflow

What Australia keeps

- AUD issuance
- ESA balances
- RITS finality
- domestic emergency law

What it could lose

- price discovery and routing
- collateral allocation
- transaction data
- operational continuity

Final settlement in public money can coexist with dependence everywhere else.

The case for conceding layers

- **Scale economics**: duplicating clearing, identity and orchestration nationally is expensive; global pools price and net better.
- **Capability scarcity**: the RBA's own partly-observed PFMI ratings show that public operation is not automatically resilient.
- **Standardisation**: common rails reduce fragmentation, error rates and compliance cost for every participant.

The efficiency case is real. It argues for shared layers on contestable terms — not for uninspectable, unsubstitutable ones. The dispute is about terms, not participation.

Sources: RBA 2026 RITS self-assessment; CPMI-IOSCO efficiency principles; companion-report counterarguments chapter.

Australia is already piloting the future stack

- **Project Acacia:** RBA and DFCRC wholesale tokenisation programme, spanning the A\$350 billion-plus repo market and other asset classes.
- CBA–J.P. Morgan (Kinexys)–ASX–HQLAx use case, with Deutsche Börse Group as HQLAx's trusted third party; completed May 2026.
- **Repo transactions settled using a CBA deposit token** alongside a pilot wholesale CBDC; the securities leg ran on HQLAx with tokenised assets held at ASX.

The domestic pilot already contains every element of this briefing: commercial-bank settlement money, a US bank's platform and a Deutsche Börse-linked collateral layer above local assets.

documented

Sources: RBA, Project Acacia; CBA, May 2026 completion; J.P. Morgan, Kinexys trial.

The tests map onto existing law

Instrument	What it reaches
Payment Systems (Regulation) Act, as amended	ministerial designation reaches orchestration layers and foreign operators
RBA financial stability / CS-facility standards	binding conditions on licensed facilities, incl. overseas licensees such as LCH
APRA CPS 230	material-service-provider obligations: the home for rehearsed substitution and exit
Security of Critical Infrastructure Act	obligations and assistance powers for designated financial assets

The machinery exists. The unfinished work is applying it deliberately to layers 4–8.

Sources: Treasury payments-licensing reforms; RBA standards; APRA CPS 230; SOCI Act.

The trusted optimiser must pass seven tests

1. Can the RBA inspect and reproduce every material decision?
2. Are liquidity locks participant-authorized and bounded?
3. Can the RBA revoke them during an emergency?
4. Is there a direct route that bypasses the optimiser?
5. Can another operator substitute without data loss?
6. Does Australia retain keys, telemetry and recovery authority?
7. Can the AUD domain isolate and continue domestically?

The settlement rule must be explicit

A net interbank obligation denominated in Australian dollars is finally discharged only in RBA money, unless the RBA expressly authorises another settlement asset for a defined use case.

- Token movement is not enough.
- Contractual netting is not enough.
- A foreign operator's commitment record is not enough.

External ESA locks need emergency limits

External ESA locks must be capped, participant-authorised and revocable under RBA emergency rules.

- Maximum amount and duration set ex ante
- Explicit participant consent for every reservation class
- RBA visibility and unilateral crisis release authority
- No permanent exclusive synchroniser
- Tested fallback to direct RITS submission

Continuity requires rehearsed exit

- Domestic copies of authoritative data and audit trails
- Source, build, key and privileged-access arrangements
- Operator substitution and manual contingency modes
- Publicly funded engineering capability inside Australia
- Crisis exercises that assume the external layer is unavailable

Exit that has never been rehearsed is not an exit plan.

The conclusion is adversarial, not conspiratorial

Retaining the last ledger entry is insufficient if private institutions control every condition that precedes it.

- Assume every actor will exploit lawful control and switching costs.
- Demand transparent optimisation and public emergency authority.
- Preserve multiple routes, domestic competence and credible substitution.
- Distinguish documented conflict from unproved coordination.

Sources: The Sovereign Ledger integrated briefing, 5 August 2026.

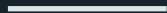
FINAL PRINCIPLE

No private owner, foreign parent, bank club or technology supplier should be able to hold the continuity of money and title hostage.

Contestable control is the operational meaning of sovereignty.

APPENDIX

Participants, ownership and sources



Reference material

Agorá participants: global banks

- BNP Paribas
- BNY
- Citi
- Crédit Agricole CIB
- Deutsche Bank
- HSBC
- JPMorgan Chase
- Standard Chartered
- UBS
- Santander
- BBVA
- CaixaBank
- Commerzbank
- Groupe BPCE
- Lloyds
- NatWest
- FNBO
- TD Bank N.A.

Sources: BIS/IIF Agorá final participant list.

Agorá participants: Japan and Korea

Japan

- Mizuho
- MUFG
- SBI Shinsei
- SMBC

Korea

- Hana
- Industrial Bank of Korea
- KB Kookmin
- NongHyup
- Shinhan
- Woori

Sources: BIS/IIF Agorá final participant list.

Agorá participants: specialist banks

Mexico / Latin America

- Banorte
- Monex
- Banco BV

Switzerland

- AMINA
- BCV
- Basler Kantonalbank

Switzerland continued

- PostFinance
- Sygnum

These institutions add regional, crypto-bank, cooperative and specialist-FX perspectives.

Sources: BIS/IIF Agorá final participant list.

Agorá participants: FMIs and networks

- **Eurex Clearing** — sole CCP
- **Euroclear** — CSD/ICSD, custody and collateral
- **SIX Digital Exchange** — regulated DLT CSD/exchange
- **Mastercard** — network rules, identity and distribution
- **Visa** — network/risk rules and bank-token platform
- **Swift** — ISO 20022, connectivity and competing ledger strategy

Sources: BIS/IIF participant list; participant corporate disclosures.

Other projects test different control points

Project	Mechanism	Control question
Mariana Cedar–Ubin+	AMM for wCBDC FX atomic links across separate ledgers	who sets curves and supplies liquidity? can interoperability avoid a central platform?
Helvetia	live wCBDC versus RTGS link	integrated token or synchronised conventional money?
Pontes	DLT-to-TARGET bridge	how does market DLT reach public money?
Meridian FX	external queue and liquidity saving	can an optimiser shape settlement before RTGS?

Sources: BIS Innovation Hub project reports; ECB Pontes; SNB Helvetia; New York Fed Cedar.

Ownership atlas: incumbent infrastructure

Institution	Ownership	Control point
Swift	financial-institution cooperative	messaging and future ledger
CLS	large financial institutions	FX acceptance and netting
TCH	large US member banks	CHIPS, RTP and token proposal
DTCC	user/industry owned	US post-trade chain
Clearstream	Deutsche Börse subsidiary	CSD/ICSD and collateral
Euroclear	mixed private, public and sovereign holders	CSD/ICSD and collateral
ASX	listed company	Australian clearing and CHES

Ownership atlas: emerging infrastructure



Institution	Ownership / governance	Control point
Digital Asset Canton Foundation	private strategic/venture coalition member/super-validator voting	Daml/Canton software and ecosystem Global Synchronizer governance
Fnality	24 banks/FMIs	reserve-backed Besu payment systems
R3	private strategic investors	Corda product and expertise
Consensys	private founder/venture group	Ethereum tools, Infura, Besu lineage
Fireblocks	private venture-backed vendor	keys, policy engine and nodes
Securitize	NYSE-listed SECZ	KYC, transfer agent, ATS and funds

Primary sources: public-money architecture

- Reserve Bank of Australia — RITS
- Bank of England — RTGS and CHAPS
- ECB — TARGET Services
- SNB — Project Helvetia
- Payments Canada — Lynx
- BIS — Project Agorá final report

Primary sources: projects and ledgers

- BIS — Project Dunbar
- BIS — Project Jura
- BIS — Project mBridge
- BIS — Agorá real-value testing
- New York Fed — Project Cedar
- Swift — 17-bank ledger cohort
- TCH — bank-led on-chain money

Primary sources: technology lineage

- Digital Asset — March 2015 launch
- Digital Asset — milestones and founders
- Digital Asset — 2016 institutional round
- Linux Foundation — Hyperledger formation
- LF Decentralized Trust — Besu
- Ethereum — official history

Primary sources: ASX and ownership

- ASIC — ASX inquiry findings
- ASIC — CHESS penalty
- Swift — cooperative governance
- CLS — shareholders
- Euroclear — major shareholders
- Deutsche Börse / Clearstream

Primary sources: Australia and clearing

- RBA — Project Acacia
- CBA — Acacia repo trial completion (May 2026)
- J.P. Morgan — Kinexys Acacia trial
- CFR — Australian OTC derivatives market reports
- LSEG — LCH SwapClear
- Indeval — DALÍ clearing and settlement